

DR ESLAM AHMED
eliteigcse.com

Binomial Expansion

Every term is $\binom{n}{r} a^{n-r} (bx)^r$. Keep the sign and the number inside the bracket with x , and every question becomes one term.

THE TOPIC ON ONE MAP

Binomial Expansion

01 Expand $(a + bx)^n$

first 4 terms

one term: $\binom{n}{r}$

simplest form

02 Find the constant

coefficient given

ratio of two coefficients

03 Multiply by a bracket

pick the pairs

term independent of x

04 Approximations

choose x

over or under?

THE GENERAL TERM

$$(a + bx)^n: \text{ term in } x^r = \binom{n}{r} a^{n-r} (bx)^r$$

powers of a go down, powers of x go up, they always add to n · $\binom{n}{r}$ on the calculator: nCr

01 Expand $(a + bx)^n$: One Term at a Time

Write each term as $\binom{n}{r} a^{n-r} (bx)^r$ and simplify it on its own line. The bracket round (bx) matters: the power applies to the number **and** the sign.

WHEN YOU SEE → YOU DO

“the first 4 terms, in ascending powers of x ”

→ $r = 0, 1, 2, 3$

“giving each term in simplest form”

→ work out every coefficient to a number (or in k)

“find the coefficient of x^7 ”

→ only the term with $r = 7$

$(2 - \frac{kx}{4})^8$

→ $b = -\frac{k}{4}$: odd powers are negative

“the 2nd, 3rd and 5th terms”

→ $r = 1, 2, 4$

WORKED MODEL • JUNE 2022

First four terms of $(2 + \frac{3}{8}x)^{10}$.

$$1 \quad 2^{10} + \binom{10}{1} 2^9 \left(\frac{3}{8}x\right) + \binom{10}{2} 2^8 \left(\frac{3}{8}x\right)^2 + \binom{10}{3} 2^7 \left(\frac{3}{8}x\right)^3$$

general term

ANSWER $1024 + 1920x + 1620x^2 + 810x^3$

IDEA 01

Practise: Expand $(a + bx)^n$: One Term at a Time

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM First four terms

P2 · TOPIC 4 3 marks

Find the first four terms, in ascending powers of x , of $(1 + 2x)^6$.

Q2 DR ESLAM One term only

P2 · TOPIC 4 2 marks

Find the coefficient of x^3 in $(2 - 3x)^5$.

$\binom{5}{3} 2^2 (-3x)^3$ — the sign goes with the x .

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

P2 · TOPIC 4 3 marks

1 Evaluate

(i) $4!$

(ii) 5C_2

(iii) 6C_3

(3 marks)

Q4**PRACTICE**

Practice · Medium

P2 · TOPIC 4 3 marks1 Expand $(2 + x)^4$.

(3 marks)

Q5**PRACTICE**

Practice · Very Hard

P2 · TOPIC 4**3 marks**

3 Given that ${}^nC_3 = 35$ find the value of n .

(3 marks)**PAST PAPERS****Q6****PAST PAPER**

Pearson IAL WMA12 · June 2022 · Q1

P2 · TOPIC 4**4 marks**

1. Find the first four terms, in ascending powers of x , of the binomial expansion of

$$\left(2 + \frac{3}{8}x\right)^{10}$$

Give each coefficient as an integer.

(4)**(Total 4 marks)**

Q7 PAST PAPER Pearson IAL WMA12 · October 2023 · Q10

P2 · TOPIC 4 12 marks

10. (i) (a) Find, in ascending powers of x , the 2nd, 3rd and 5th terms of the binomial expansion of

$$(3 + 2x)^6 \quad (3)$$

For a particular value of x , these three terms form consecutive terms in a geometric series.

- (b) Find this value of x . (3)

- (ii) In a **different** geometric series,

- the first term is $\sin^2 \theta$
- the common ratio is $2 \cos \theta$
- the sum to infinity is $\frac{8}{5}$

- (a) Show that

$$5 \cos^2 \theta - 16 \cos \theta + 3 = 0 \quad (3)$$

- (b) Hence find the exact value of the 2nd term in the series. (3)

(Total for Question 10 is 12 marks)

Q8 **PAST PAPER** Pearson IAL WMA12 · January 2024 · Q2**P2 · TOPIC 4** **3 marks**

2. Find the coefficient of the term in x^7 of the binomial expansion of

$$\left(\frac{3}{8} + 4x\right)^{12}$$

giving your answer in simplest form.

(3)

(Total for Question 2 is 3 marks)

02 Find the Unknown Constant

Write the terms with the constant still in them, then turn the sentence into an equation: “the coefficient is 540”, “ $B = 18D$ ”, “the coefficient of x^2 is 3 times the coefficient of x^3 ”.

WHEN YOU SEE → YOU DO

“the first three terms are $1, -4x, px^2$ ”

→ match term by term: first find k , then p

“one of the terms is $540x^4$ ”

→ $\binom{6}{4}3^2a^4 = 540$

“coefficient of x^2 is 3 times coefficient of x^3 ”

→ one equation, cancel common powers of k

“the coefficients form an arithmetic progression”

→ $2 \times \text{middle} = \text{first} + \text{last}$

“when $f(x)$ is divided by $(4x - 5)$ the remainder is ...”

→ remainder theorem: $f(\frac{5}{4})$

WORKED MODEL • JANUARY 2026

$(3 + kx)^7$: the coefficient of x^2 is 3 times the coefficient of x^3 . Find k .

1 x^2 : $5103k^2$ x^3 : $2835k^3$

the two terms

2 $5103k^2 = 3 \times 2835k^3$

the sentence as an equation

ANSWER $k = \frac{5103}{8505} = \frac{3}{5}$

IDEA 02

Practise: Find the Unknown Constant

My questions first, then selected practice, then real past papers.

Q9**DR ESLAM**

Coefficient given

P2 · TOPIC 4 3 marks

In the expansion of $(1 + kx)^8$, $k > 0$, the coefficient of x^2 is 112. Find k .

SELECTED PRACTICE**Q10****PRACTICE**

Practice · Easy

P2 · TOPIC 4 3 marks

7 In the expansion of $(p + x)^{12}$, the coefficient of the x^5 term is 12 976 128. Find the value of p .

(3 marks)

Q11**PRACTICE**

Practice · Hard

P2 · TOPIC 4**3 marks**

- 6 In the expansion of $(3a - 2x)^6$, the coefficient of the x^3 term is equal to the coefficient of the x^4 term. Find the value of a .

(3 marks)**Q12****PRACTICE**

Practice · Very Hard

P2 · TOPIC 4**5 marks**

- 10 In the expansion of $(a + bx)^8$, the coefficient of the x^5 term is -870 912.
In the expansion of $(a + bx)^{12}$, the coefficient of the x^3 term is -1 557 135 360.

Find the possible values of a and b .

(5 marks)

PAST PAPERS

Q13 PAST PAPER Pearson IAL WMA12 · January 2020 · Q2

P2 · TOPIC 4 7 marks

2. One of the terms in the binomial expansion of $(3 + ax)^6$, where a is a constant, is $540x^4$

(a) Find the possible values of a .

(4)

(b) Hence find the term independent of x in the expansion of

$$\left(\frac{1}{81} + \frac{1}{x^6}\right)(3 + ax)^6$$
(3)

(Total 7 marks)

Q14 PAST PAPER Pearson IAL WMA12 · October 2021 · Q1

P2 · TOPIC 4 6 marks

1. The first three terms, in ascending powers of x , of the binomial expansion of $(1 + kx)^{16}$ are

$$1, -4x \text{ and } px^2$$

where k and p are constants.

- (a) Find, in simplest form,

(i) the value of k

(ii) the value of p

(3)

$$g(x) = \left(2 + \frac{16}{x}\right)(1 + kx)^{16}$$

Using the value of k found in part (a),

- (b) find the term in x^2 in the expansion of $g(x)$.

(3)

(Total 6 marks)

Q15 PAST PAPER Pearson IAL WMA12 · October 2022 · Q2

P2 · TOPIC 4 7 marks

2. A curve C has equation $y = f(x)$ where

$$f(x) = (2 - kx)^5$$

and k is a constant.

Given that when $f(x)$ is divided by $(4x - 5)$ the remainder is $\frac{243}{32}$

- (a) show that $k = \frac{2}{5}$

(2)

- (b) Find the first three terms, in ascending powers of x , of the binomial expansion of

$$\left(2 - \frac{2}{5}x\right)^5$$

giving each term in simplest form.

(3)

Using the solution to part (b) and making your method clear,

- (c) find the gradient of C at the point where $x = 0$

(2)

(Total for Question 2 is 7 marks)

Q16**PAST PAPER**

Pearson IAL WMA12 · January 2023 · Q3

P2 · TOPIC 4 7 marks

3.
$$f(x) = \left(2 + \frac{kx}{8}\right)^7$$
 where k is a non-zero constant

- (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of $f(x)$.
Give each term in simplest form.

(4)

Given that, in the binomial expansion of $f(x)$, the coefficients of x , x^2 and x^3 are the first 3 terms of an arithmetic progression,

- (b) find, using algebra, the possible values of k .

(Solutions relying entirely on calculator technology are not acceptable.)

(3)

(Total for Question 3 is 7 marks)

Q17 **PAST PAPER** Pearson IAL WMA12 · June 2023 · Q4**P2 · TOPIC 4** **7 marks**

4. The binomial expansion, in ascending powers of x , of

$$(3 + px)^5$$

where p is a constant, can be written in the form

$$A + Bx + Cx^2 + Dx^3 \dots$$

where A , B , C and D are constants.

- (a) Find the value of A

(1)

Given that

- $B = 18D$
- $p < 0$

- (b) find

- (i) the value of p
- (ii) the value of C

(6)

(Total for Question 4 is 7 marks)

Q18 **PAST PAPER** Pearson IAL WMA12 · October 2025 · Q2**P2 · TOPIC 4** **6 marks**

2. The first 4 terms, in ascending powers of x , in the binomial expansion of

$$(1 + px)^{10}$$

are

$$1 + 15x + qx^2 + rx^3$$

where p , q and r are constants.

Find the value of p , the value of q and the value of r .

(6)

(Total for Question 2 is 6 marks)

Q19**PAST PAPER**

Pearson IAL WMA12 · January 2026 · Q1

P2 · TOPIC 4 6 marks

1. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$(3 + kx)^7$$

where k is a non-zero constant. Give each term in simplest form.

(4)

Given that, in this expansion, the coefficient of x^2 is 3 times the coefficient of x^3

- (b) find, in simplest form, the value of k .

(2)

(Total for Question 1 is 6 marks)

03 Multiply by a Bracket: Pick the Pairs

For $(p + qx)(\text{expansion})$ you only need a few products: to get x^2 , pair p with the x^2 term and qx with the x term. For a term in $\frac{1}{x}$ or $\frac{1}{x^6}$, pair it with the term that cancels the power.

WHEN YOU SEE → YOU DO

“the coefficient of x^2 in $(3 - \frac{1}{2}x)(2 + px)^6$ ”

→ $3 \times (x^2 \text{ term}) - \frac{1}{2} \times (x \text{ term})$

“the term independent of x ”

→ powers must add to 0

$(\frac{x^2+8}{x^5})(1 + \frac{x}{4})^{12}$

→ split: x^{-3} and $8x^{-5}$, pair with x^3 and x^5

“the constant term is 3 times the coefficient of x ”

→ two pair-sums, one equation

WORKED MODEL • JUNE 2024

Coefficient of x^3 in $(10x + 3)(1 - \frac{1}{6}x)^9$, given $1 - \frac{3}{2}x + x^2 - \frac{7}{18}x^3$.

1 $10x \times x^2$ and $3 \times (-\frac{7}{18}x^3)$

pick the pairs

ANSWER $10 - \frac{7}{6} = \frac{53}{6}$

× **TRAP** Multiplying out every term. Only the pairs that make the power you want matter.

IDEA 03

Practise: Multiply by a Bracket: Pick the Pairs

My questions first, then selected practice, then real past papers.

Q20 **DR ESLAM** Pick the pairs

P2 · TOPIC 4 3 marks

Find the coefficient of x^2 in $(2 + x)(1 - x)^6$.

SELECTED PRACTICE**Q21** **PRACTICE** Practice · Medium

P2 · TOPIC 4 6 marks

8 (a) Find the first three terms in the expansion of $(3 + 2x)^8$.

(3 marks)

(b) Given that x is small such that x^3 and higher powers of x can be ignored show that

$$(1 + x)(3 + 2x)^8 \approx 6561 + 41553x + 116640x^2$$

(3 marks)

Q22**PRACTICE**

Practice · Hard

P2 · TOPIC 4 6 marks

7 (a) Find the first three terms in the expansion of $(2 - 3x)^7$.

(3 marks)

(b) Given that x is small such that x^3 and higher powers of x can be ignored show that
 $(1 - 2x)(2 - 3x)^7 \approx 128 - 1600x + 8736x^2$

(3 marks)

Q23**PRACTICE**

Practice · Very Hard

P2 · TOPIC 4 6 marks

7 (a) Find the first three terms in the expansion of $(4 - 3x)^9$.

(3 marks)

(b) Given that x is small such that x^3 and higher powers of x can be ignored show that

$$(3 - 2x^2)(4 - 3x)^9 \approx 786432 - 5308416x + 15400960x^2$$

(3 marks)

PAST PAPERS

Q24 PAST PAPER Pearson IAL WMA12 · October 2019 · Q3

P2 · TOPIC 4 6 marks

3. (a) Find the first 4 terms, in ascending powers of x , in the binomial expansion of

$$\left(1 + \frac{x}{4}\right)^{12}$$

giving each coefficient in its simplest form.

(3)

- (b) Find the term independent of x in the expansion of

$$\left(\frac{x^2 + 8}{x^5}\right)\left(1 + \frac{x}{4}\right)^{12}$$

(3)

(Total 6 marks)

Q25 **PAST PAPER** Pearson IAL WMA12 · January 2021 · Q4**P2 · TOPIC 4** **8 marks**

4. (a) Find the first three terms, in ascending powers of x , of the binomial expansion of

$$(2 + px)^6$$

where p is a constant. Give each term in simplest form.

(4)

Given that in the expansion of

$$\left(3 - \frac{1}{2}x\right)(2 + px)^6$$

the coefficient of x^2 is $-\frac{3}{4}$

- (b) find the possible values of p .

(4)

(Total 8 marks)

Q26 **PAST PAPER** Pearson IAL WMA12 · January 2022 · Q3**P2 · TOPIC 4** **7 marks**

3. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$\left(2 - \frac{kx}{4}\right)^8$$

where k is a non-zero constant. Give each term in simplest form.

(4)

$$f(x) = (5 - 3x)\left(2 - \frac{kx}{4}\right)^8$$

In the expansion of $f(x)$, the constant term is 3 times the coefficient of x .

- (b) Find the value of k .

(3)

(Total 7 marks)

Q27**PAST PAPER**

Pearson IAL WMA12 · June 2024 · Q1

P2 · TOPIC 4 5 marks

1. (a) Find the first four terms, in ascending powers of x , of the binomial expansion of

$$\left(1 - \frac{1}{6}x\right)^9$$

giving each term in simplest form.

(3)

- (b) Hence find the coefficient of x^3 in the expansion of

$$(10x + 3)\left(1 - \frac{1}{6}x\right)^9$$

giving the answer in simplest form.

(2)

(Total for Question 1 is 5 marks)

Q28**PAST PAPER**

Pearson IAL WMA12 · June (R paper) 2024 · Q4

P2 · TOPIC 4 7 marks

4. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$(3 + 2x)^6$$

giving each coefficient in simplest form.

(4)

- (b) Hence find the coefficient of x^2 in the expansion of

$$\left(2x^2 - \frac{1}{6x}\right)(3 + 2x)^6$$

(3)

(Total for Question 4 is 7 marks)

Q29 **PAST PAPER** Pearson IAL WMA12 · October 2024 · Q5**P2 · TOPIC 4** **7 marks**

5. (a) Find, in terms of a , the first 3 terms, in ascending powers of x , of the binomial expansion of

$$(2 + ax)^6$$

where a is a non-zero constant. Give each term in simplest form.

(3)

$$f(x) = \left(3 + \frac{1}{x}\right)^2 (2 + ax)^6$$

Given that the constant term in the expansion of $f(x)$ is 576

- (b) find the value of a .

(4)

(Total for Question 5 is 7 marks)

Q30 **PAST PAPER** Pearson IAL WMA12 · June 2025 · Q1**P2 · TOPIC 4** **6 marks**

1: (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$(1 - 4x)^7$$

giving each term in simplest form.

(4)

In the series expansion of

$$(5 + kx)(1 - 4x)^7 \quad \text{where } k \text{ is a constant}$$

the coefficient of the term in x^2 is 1316

(b) Use the answer to part (a) to find the value of k .

(2)

(Total for Question 1 is 6 marks)

04 Approximations: Choose the Right x

Match the bracket to the number you want: $(2 + \frac{x}{8})^{13}$ with $2 + \frac{x}{8} = 2.0125$ gives $x = 0.1$. Substitute into the expansion. Small x keeps the ignored terms tiny; state over- or under-estimate from the signs of those terms.

WHEN YOU SEE → YOU DO

“use your expansion to find an approximation for 2.0125^{13} ”

→ solve $2 + \frac{x}{8} = 2.0125$ for x

“state the value of x that should be used”

→ just that equation — no calculation

“overestimate or underestimate?”

→ the missing terms are positive → underestimate

“show that an estimate for $(\frac{33}{32})^{10}$ is $\frac{1389}{k}$ ”

→ $1 + \frac{5}{16}x = \frac{33}{32} \Rightarrow x = \frac{1}{10}$

WORKED MODEL • JUNE 2026

$(1 + \frac{5}{16}x)^{10} \approx 1 + \frac{25}{8}x + \frac{1125}{256}x^2$. Estimate $(\frac{33}{32})^{10}$.

1 $x = \frac{1}{10}$: $1 + \frac{5}{16} + \frac{45}{1024}$

substitute

ANSWER $\frac{1389}{1024}$, $k = 1024$

IDEA 04

Practise: Approximations: Choose the Right x

My questions first, then selected practice, then real past papers.

Q31 DR ESLAM Choose x

P2 · TOPIC 4 3 marks

Expand $(1 + 2x)^5$ up to x^2 and use it to estimate 1.02^5 .

SELECTED PRACTICE**Q32** PRACTICE Practice · Easy

P2 · TOPIC 4 5 marks

8 (a) Find the first three terms in the expansion of $(5 + 2x)^5$.

(3 marks)

(b) Use your answer to part (a) to estimate the value of $(5.04)^5$.

(2 marks)

Q33**PRACTICE**

Practice · Medium

P2 · TOPIC 4 5 marks

5 (a) Find the first three terms in the expansion of $(9 - 2x)^5$.

(3 marks)

(b) Use your answer to part (a) to estimate $(8.9)^5$.

(2 marks)

Q34 **PRACTICE** Practice · Very Hard**P2 · TOPIC 4** 7 marks

4 (a) Use the first three terms, in ascending powers of x , in the expansion of $(3 - 5x)^4$ to find an approximation for $(2.6)^4$.

(5 marks)

(b) Using your calculator, find the percentage error in the approximation from part (a) to the exact value of $(2.6)^4$.

(2 marks)

PAST PAPERS

Q35 PAST PAPER Pearson IAL WMA12 · June 2021 · Q4

P2 · TOPIC 4 8 marks

4. (a) Find, in ascending powers of x , up to and including the term in x^3 , the binomial expansion of

$$\left(2 + \frac{x}{8}\right)^{13}$$

fully simplifying each coefficient.

(4)

- (b) Use the answer to part (a) to find an approximation for 2.0125^{13}

Give your answer to 3 decimal places.

(3)

Without calculating 2.0125^{13}

- (c) state, with a reason, whether the answer to part (b) is an overestimate or an underestimate.

(1)

(Total 8 marks)

Q36 **PAST PAPER** Pearson IAL WMA12 · January 2025 · Q2**P2 · TOPIC 4** **5 marks**

2. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$(2 - 5x)^8$$

giving each term in simplest form.

(4)

This expansion is to be used to find an approximation for 2.05^8

- (b) State the value of x that should be used.

(There is no need to carry out this calculation.)

(1)

(Total for Question 2 is 5 marks)

Q37 PAST PAPER Pearson IAL WMA12 · June 2026 · Q7

P2 · TOPIC 4 9 marks

7. (a) Find, in ascending powers of x , up to and including the term in x^2 , the binomial expansion of

$$\left(1 + \frac{5}{16}x\right)^{10}$$

fully simplifying each coefficient.

(4)

- (b) Use $x = \frac{1}{10}$ and the answer to part (a) to show that an estimate for $\left(\frac{33}{32}\right)^{10}$ is

$$\frac{1389}{k}$$

where k is an integer to be found.

(3)

- (c) Hence find a rational estimate for $\sqrt[10]{1389}$

(2)

(Total for Question 7 is 9 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|------------------------|--|
| 01 | Expand | $\binom{n}{r}a^{n-r}(bx)^r$; brackets round (bx) . |
| 02 | Constants | Terms with k in; the sentence becomes an equation. |
| 03 | Times a bracket | Pick only the pairs that make the power. |
| 04 | Approximations | Bracket = the number $\rightarrow x$; sign of the next terms. |

BOTTOM LINE One general term • keep the sign with x • pick the pairs • choose x from the bracket.

Answers

Binomial Expansion · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · EXPAND $(a + bx)^n$: ONE TERM AT A TIME

Q1	Dr Eslam · First four terms	$1 + 12x + 60x^2 + 160x^3$
Q2	Dr Eslam · One term only	$10 \times 4 \times (-27) = -1080$
Q3	Practice · Easy	(i) 24 (ii) 10 (iii) 20
Q4	Practice · Medium	$16 + 32x + 24x^2 + 8x^3 + x^4$
Q5	Practice · Very Hard	$\frac{n(n-1)(n-2)}{6} = 35 \Rightarrow n = 7$
Q6	Pearson IAL WMA12 · June 2022 · Q1	$1024 + 1920x + 1620x^2 + 810x^3$
Q7	Pearson IAL WMA12 · October 2023 · Q10	(i)(a) $2916x, 4860x^2, 2160x^4$. (i)(b) $x = \frac{15}{4}$. (ii)(a) Proven. (ii)(b) $\frac{48}{125}$
Q8	Pearson IAL WMA12 · January 2024 · Q2	96228

IDEA 02 · FIND THE UNKNOWN CONSTANT

- Q9** Dr Eslam · Coefficient given $28k^2 = 112 \Rightarrow k = 2$
- Q10** Practice · Easy $\binom{12}{5}p^7 = 12\,976\,128 \Rightarrow p^7 = 16384: p = 4$
- Q11** Practice · Hard $a = -\frac{1}{2}$
- Q12** Practice · Very Hard $a = 4, b = -3$ or $a = -4, b = 3$
- Q13** Pearson IAL WMA12 · January 2020 · Q2 $(a) a = \pm\sqrt{2}. (b) 17$
- Q14** Pearson IAL WMA12 · October 2021 · Q1 $(a)(i) k = -\frac{1}{4}. (a)(ii) p = \frac{15}{2}. (b) -125x^2$
- Q15** Pearson IAL WMA12 · October 2022 · Q2 $(a) k = \frac{2}{5}. (b) 32 - 32x + \frac{64}{5}x^2. (c) -32$
- Q16** Pearson IAL WMA12 · January 2023 · Q3 $(a) 128 + 56kx + \frac{21}{2}k^2x^2 + \frac{35}{32}k^3x^3. (b) k = \frac{16}{5}$
or $k = 16$
- Q17** Pearson IAL WMA12 · June 2023 · Q4 $(a) A = 243. (b)(i) p = -\frac{1}{2}. (b)(ii) C = \frac{135}{2}$
- Q18** Pearson IAL WMA12 · October 2025 · Q2 $p = \frac{3}{2}, q = \frac{405}{4}, r = 405$
- Q19** Pearson IAL WMA12 · January 2026 · Q1 $(a) 2187 + 5103kx + 5103k^2x^2 + 2835k^3x^3. (b) k = \frac{3}{5}$

IDEA 03 · MULTIPLY BY A BRACKET: PICK THE PAIRS

- Q20** Dr Eslam · Pick the pairs $(1 - x)^6 = 1 - 6x + 15x^2 \dots: 2(15) + 1(-6) = 24$
- Q21** Practice · Medium (a) $6561 + 34992x + 81648x^2$ (b) multiply by $(1 + x)$, keep up to x^2
- Q22** Practice · Hard (a) $128 - 1344x + 6048x^2$ (b) multiply by $(1 - 2x)$, keep up to x^2
- Q23** Practice · Very Hard (a) $262144 - 1769472x + 5308416x^2$ (b) multiply by $(3 - 2x^2)$, keep up to x^2
- Q24** Pearson IAL WMA12 · October 2019 · Q3 (a) $1 + 3x + \frac{33}{8}x^2 + \frac{55}{16}x^3$. (b) $\frac{77}{8}$
- Q25** Pearson IAL WMA12 · January 2021 · Q4 (a) $64 + 192px + 240p^2x^2$. (b) $p = \frac{1}{8}$ or $p = \frac{1}{120}$
- Q26** Pearson IAL WMA12 · January 2022 · Q3 (a) $256 - 256kx + 112k^2x^2 - 28k^3x^3$. (b) $k = -\frac{14}{15}$
- Q27** Pearson IAL WMA12 · June 2024 · Q1 (a) $1 - \frac{3}{2}x + x^2 - \frac{7}{18}x^3$. (b) $\frac{53}{6}$
- Q28** Pearson IAL WMA12 · June (R paper) 2024 · Q4 (a) $729 + 2916x + 4860x^2 + 4320x^3$. (b) 738
- Q29** Pearson IAL WMA12 · October 2024 · Q5 (a) $64 + 192ax + 240a^2x^2$. (b) $a = -\frac{24}{5}$
- Q30** Pearson IAL WMA12 · June 2025 · Q1 (a) $1 - 28x + 336x^2 - 2240x^3$. (b) $k = 13$

IDEA 04 · APPROXIMATIONS: CHOOSE THE RIGHT x

- Q31** Dr Eslam · Choose x $1 + 10x + 40x^2$; $x = 0.01$: 1.104
- Q32** Practice · Easy (a) $3125 + 6250x + 5000x^2$ (b) $x = 0.02$: 3252
- Q33** Practice · Medium (a) $59049 - 65610x + 29160x^2$ (b) $x = 0.05$: 55841.4
- Q34** Practice · Very Hard (a) $81 - 540x + 1350x^2$, $x = 0.08$: 46.44 (b) exact 45.6976: error $\approx 1.62\%$
- Q35** Pearson IAL WMA12 · June 2021 · Q4 (a) $8192 + 6656x + 2496x^2 + 572x^3$. (b) 8883.132. (c) Underestimate
- Q36** Pearson IAL WMA12 · January 2025 · Q2 (a) $256 - 5120x + 44800x^2 - 224000x^3$. (b) $x = -0.01$
- Q37** Pearson IAL WMA12 · June 2026 · Q7 (a) $1 + \frac{25}{8}x + \frac{1125}{256}x^2$ (b) $1 + \frac{5}{16} + \frac{45}{1024} = \frac{1389}{1024}$.
 $k = 1024$ (c) $\frac{33}{16}$